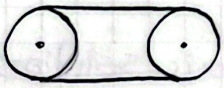
A9 Pulley Design

Given:

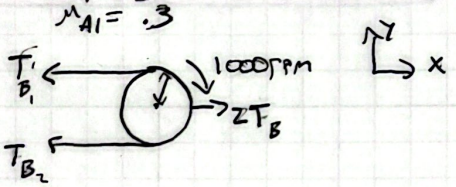
- $\varnothing_p = 3$ in
- RPM = 1000 rpm
- $T_B = 4$ lbf
- Pulley material = Aluminium
- SF = 3
- Belt material = rubber
  $S_{y_B} = 3.25$ Mpa (471 psi) at 20% strain
- L = 6 in



1.1. Belt Selection:

- Knowns: $\varnothing_p = 3$ in
  RPM = 1000 rpm
  $T_B = 4$ lbf
  $S_y = 471$ psi at 20% strain
  SF = 3
  $\mu_{Al} = .3$
- Unknown: $\varnothing_B$
  F_1
  F_2
  μ

- FBD:



Forces on tight and loose ends:

- Algebraic Solution:

$$P = T_w = r_w C (F_{tight} - F_{loose})$$

$$T_{B_1} = T_{B_2} = T_B = 4 \text{ lbf}$$

$$\frac{F_1}{F_2} = e^{\mu_{Al}\theta}$$

$$T_1 + T_2 = 2T$$

$$T_2 e^{\mu_{Al}} + T_2 = 2T$$

$$T_2 (e^{\mu_{Al}\theta} + 1) = 2T$$

$$\boxed{T_2 = \frac{2T}{e^{\mu_{Al}\theta} + 1}}$$

$$\boxed{T_1 = 2T - T_2}$$

- Numeric Solutions:



$$T_2 = \frac{2(4)}{e^{.3\pi} + 1}$$

$$\boxed{T_{loose_2} = 2.24 \text{ lbf}}$$

$$T_1 = 2(4) - 2.24$$

$$\boxed{T_{tight} = 5.76 \text{ lbf}}$$

Diameter of belt:

- Algebraic solutions:

$$\frac{S_y}{SF} = \frac{T_B}{A}$$

$$\frac{S_y}{SF} = \frac{T_B}{\frac{\pi}{4} D^2}$$

$$\boxed{D = \sqrt{\frac{4 T_B SF}{\pi S_y}}}$$

- Numerical solution:

$$D_B = \sqrt{\frac{4(4)(3)}{\pi(471)}}$$

$$\boxed{\phi_B = .1801 \text{ in}}$$

$$A_B = \frac{\pi}{4} D_B^2 = \frac{\pi}{4}(.1801)^2 = \boxed{.0255 \text{ in}}$$

Outer Circumference of Belt:

- Algebraic Solutions:

$$\boxed{C_B = 2L + \pi \phi_p}$$

- Numeric solutions

$$C_B = 2(6) + \pi(3)$$

$$\boxed{C_B = 21.42 \text{ in}}$$

Belt Selection:

$\phi_B = .1801 \text{ in} \approx 3/16 \text{ in}$

$C_B = 21.42 \text{ in} \approx$

$\phi_p = 3 \text{ in}$

Round Belt: 3044K605 (McMaster-Carr)
Diameter = 3/16"
Circumference = 21 1/2"

2. Shaft Diameter:

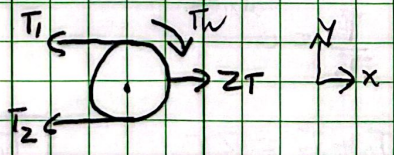
- Knowns: $M_a T = T_{right}(\phi_p/2)$

- $M_a T = T_{right}(\phi_p/2)$
- $F_2 = 2T$
- RPM = 2000 rpm
- material = 1060 HS ($E = 11600 ksi$, $S_y = 98.2 ksi$)

- Unknowns:

ϕ_s
W_p
T_{max}

- FBD:



- Algebraic Solutions:

$$\text{Torque} = F_{right}\left(\frac{\phi_r}{2}\right)$$

$$\tau = \frac{T}{J} = \frac{T\left(\frac{\phi_s}{2}\right)}{\frac{\pi \phi_s^4}{32}}$$

$$\phi_s = \sqrt[3]{\frac{16T}{\pi \tau}}$$

$$\tau = .577(S_y)$$

$$\tau = .577(98.2)$$

- Numeric Solutions:

$$T = (5.76)\left(\frac{3}{2}\right) = 8.64 \text{ in lbf}$$

$$\phi_s = \sqrt[3]{\frac{16(8.64)}{\pi(56661.4)}}$$

$E, 57Z(98.2)$

$\tau = 56.66 \text{ ksi}$

$\tau = 56661.4 \text{ psi}$

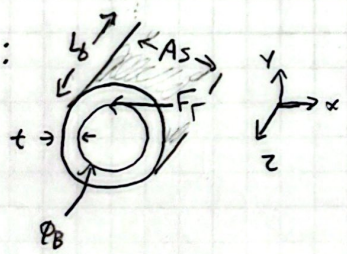
$$\phi_s = .0919 \text{ in} \approx .125 \text{ in}$$

3. Journal Bearing:

- Known:
  - Material = Sintered metal or nonmetallic
  - RPM = 2000 rpm
  - $\phi_s = .125$ in
  - RC5 Fit = hole +.7 thou
  - $L_B = 1$ in

- Unknown:
  - P
  - V
  - PV
  - ϕ_B
  - F_r
  - F_r

FBD:



- Algebraic Solution:

$$P = \frac{F_r}{A_s} = \frac{2T/\phi_B}{L_B \pi \phi_B} = \frac{2T}{L_B \pi \phi_B^2}$$

$$V = rpm\left(\frac{\pi \phi_B}{1 rev}\right)\left(\frac{1 ft}{12 in}\right)$$

Numeric Solutions:

$$P = \frac{2(8.64)}{1\pi(.1257)^2}$$

$\phi_B = .125 + .0007 = .1257$

$$P = 348.12 \text{ psi}$$

$$V = 1000\left(\frac{\pi(.1257)}{1}\right)\left(\frac{1}{12}\right)$$

$$V = 32.91 \text{ ft/min}$$

$$PV = 348.12(32.91)$$

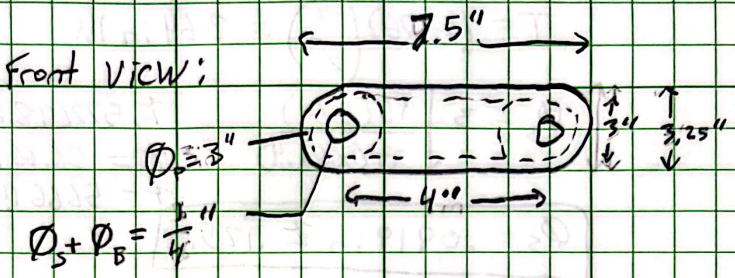
$$PV = 11456.63 \text{ psi-ft/min}$$

- Material Selection (Table 4 Machinery Handbook):

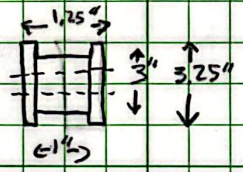
Porous Bronze

- Frame Design:

Front view:



Side view:



3D view:

